Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
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Department of Computer Science & Engineering (AI & ML)

UNISCOUT: College Recommendation System

- **Theme –** SDG 4: Quality Education
- **Category- Software**
- **Group No - 2**
- **Group Members – Vedant Kirtane , Kanhaiya Mishra, Nitin Maurya , Hemang Manjrekar.**
- **Guide- Prof . Adesh Hardas**

Outline

- Introduction
- Literature Survey
- Limitations of existing systems
- Problem statement and Objectives
- Proposed Methodology
- Experimental Setup
- Implementation
- Results & Analysis
- Conclusion and Future work
- References

Introduction

- **UNISCOUT College Recommendation System – Background**
- Engineering admissions in Maharashtra are primarily based on **MHT CET scores and CAP rounds**
- Students face difficulty in selecting the right college due to:
  - Large number of colleges and branches
  - Complex cutoff data across multiple CAP rounds
- Traditional methods involve:
  - Manual research on websites
  - Consulting unofficial sources
- There is a need for a **data-driven, automated recommendation system**
- UNISCOUT uses **real-time cutoff data and student inputs** to suggest suitable colleges

Literature Survey

Survey of Existing Solutions

- Existing systems include:
  - College prediction tools based on cutoff ranks
  - Basic filtering systems on admission portals
  - Generic recommender systems

Techniques Used

- Collaborative Filtering
- Content-Based Filtering
- Data Mining & Machine Learning
- Cutoff Trend Analysis

Limitations Identified

- Lack of personalization
- No accurate admission probability prediction
- Limited use of multi-round CAP data
- Low accuracy due to basic models
- Scattered and non-uniform datasets

Literature Survey Summary Table

Author & Year [ ]	Technique Used	Dataset Used	Key Results	Limitations
Patil et al., 2018 [1]	Data Mining	Academic dataset	Improved filtering	No personalization
Burke, 2002 [2]	Hybrid Recommender	User data	Better recommendations	Complex implementation
Su & Khoshgoftaar, 2009 [3]	Collaborative Filtering	User-item data	Accurate predictions	Cold-start problem
Adomavicius et al., 2005 [4]	Recommender Systems	Large datasets	High efficiency	Scalability issues

Limitations of existing Systems

- Lack of **personalized recommendations** based on individual student profiles
- Dependence on **manual analysis of cutoff data**, which is time-consuming
- Limited use of **historical CAP round trends** for prediction
- Absence of **admission probability estimation**
- Data is often **scattered across multiple sources**, reducing accessibility
- Existing systems provide **generic suggestions**, not tailored results
- Low accuracy due to **basic filtering techniques**
- No integration of **real-time updates**

Problem statement and Objectives

- Students face difficulty in selecting suitable colleges during the **MHT CET CAP admission process** due to:
- Large number of colleges and branches
- Complex and dynamic cutoff trends across multiple rounds
- . Existing methods rely on **manual research and scattered data sources**, leading to:
- Confusion and wrong decision-making
- Lack of personalized guidance
- There is no system that provides:
- Accurate prediction of admission chances
- Recommendations based on individual student profile

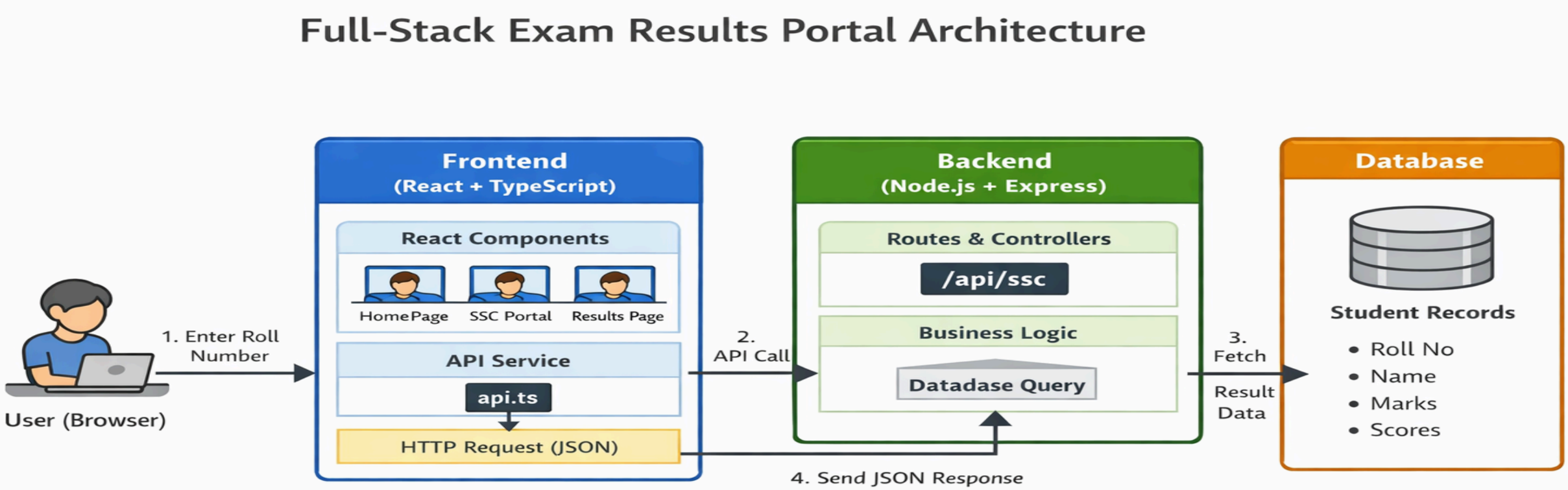
Proposed Methodology

Techniques & Model:

- Cutoff trend analysis (CAP rounds)
- Admission Probability = (Score – Min Cutoff) / (Range)
- Metrics: Accuracy, Precision, Recall, RMSE

Database & Platform:

- Tables: Student, College, Cutoff
- Web-based system (Python, Flask/Django, MySQL)



Experimental Setup

Dataset & Input Details

- **Input Data**
- Student Inputs:
  - MHT CET Score / Percentile
  - Category (Open, OBC, SC, ST, etc.)
  - Preferred Branch (CSE, IT, Mechanical, etc.)
  - Location Preference

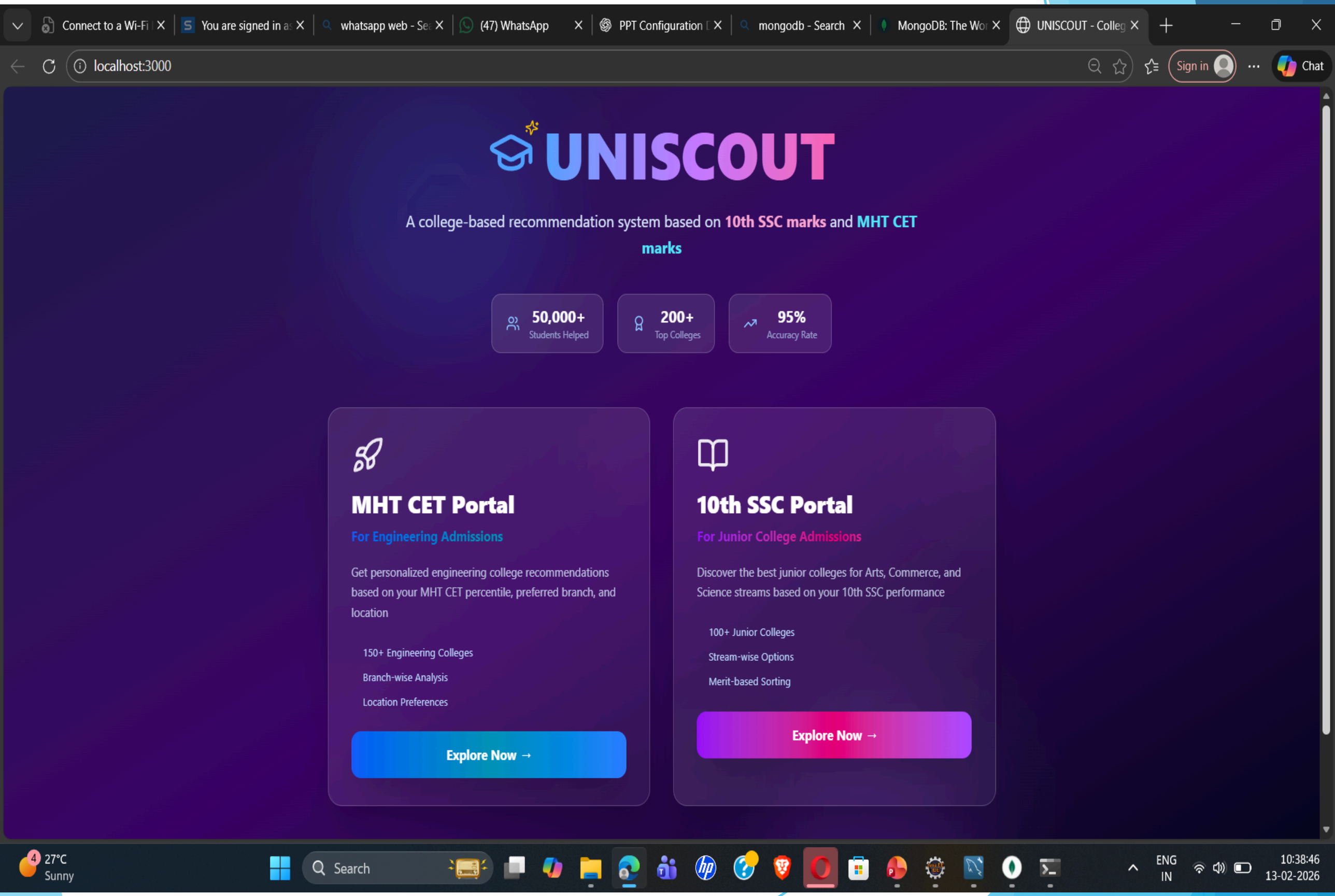
Dataset Used

- Historical **CAP Round cutoff data (MHT CET)**
- College details:
  - Branch-wise cutoffs
  - College rankings
  - Seat availability

Data Source

- Official Government Admission Portal (MHT CET Cell)
- Structured and preprocessed dataset for analysis
- **Technologies to be Used**
- Frontend: **HTML , CSS , JAVASCRIPT , REACT , CSS TAILWIND**
- Backend: **Node.js , Express.js , JSON Database , Rest API**
- Database: **MySQL/EXCEL SHEET**

Implementation



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10th SSC Portal

Discover the best junior colleges for your future

Fill in your details to get personalized recommendations

Total Marks (Out of 500)

e.g., 450

Maximum Marks

500

Academic Year

2025

Regular Round Number

Round 1 | Round 2 | Round 3

Category

Open

Stream Preference

Science | Commerce | Arts

Preferred Location

Select your preferred location

Medium of Instruction

English | Marathi

Find My Perfect Colleges

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Back

Progress 0%

MHT CET Portal

Find your perfect engineering college

Fill in your details to get personalized recommendations

Enter Your Percentile

e.g., 95.5

Academic Year

2025

CAP Round Number

Round I | Round II

Category

Open

Branch Preference

Select your preferred branch

Preferred Location

Select your preferred location

Find My Perfect Colleges

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100 Total Matches | 100 High Chance | 0 Medium Chance | 0 Low Chance

Filters | Sort by Chance | Showing 100 of 100 colleges

K. P. B. Hinduja College Of Commerce
Commerce • Mumbai
85.60% +5.60%
College Code: JC9416
Category: Open
Medium: English
Region: Sheet1
High Admission Chance
Your percentage is above the cutoff. Strong chances!

Swami Vivekanand High School & Jr College, Chembur
Commerce • Mumbai
85.60% +5.60%

S. K. Somaiya Vinay Mandir High School Jr College
Commerce • Mumbai
85.40% +5.80%

Podar International Higher Secondary School
Commerce • Mumbai
85.20% +6.00%

St Andrew's College Of Arts Sci & Com
Commerce • Mumbai
85.20% +6.00%

The B.s.g.d's Jr College Of Com,arts & Science
Commerce • Mumbai
85.20% +6.00%

Elphinstone College
Commerce • Mumbai
84.20% +7.00%

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20 Total Colleges | 19 High Chance | 1 Medium Chance | 0 Low Chance

Filters & Sort

St. Francis Institute Of Technology — High — Mumba.
5.0
Branch: Computer Science and Engineering | Cutoff: 93.0
Private | +5.0 %ile
Click for more details

Atharva College Of Engineering — High — Mumba.
5.0
Branch: Computer Science and Engineering | Cutoff: 91.7
Private | +6.3 %ile
Click for more details

Kj Somaiya College Of Engineering — High — Mumba.
5.0
Branch: Computer Science and Engineering | Cutoff: 91.1
Private | +6.9 %ile
Click for more details

Rajiv Gandhi Institute Of Technology — High — Mumba.
5.0
Branch: Computer Science and Engineering | Cutoff: 90.7
Private | +7.3 %ile
Click for more details

Atharva College Of Engineering — High — Mumba.
5.0
Branch: Computer Science and Engineering | Cutoff: 89.4
Private | +8.6 %ile
Click for more details

Rajiv Gandhi Institute Of Technology — High — Mumba.
5.0
Branch: Computer Science and Engineering | Cutoff: 87.2
Private | +10.8 %ile
Click for more details

Thakur College Of Engineering And Technology — High — Mumba.

Kj Somaiya College Of Engineering — High — Mumba.
5.0

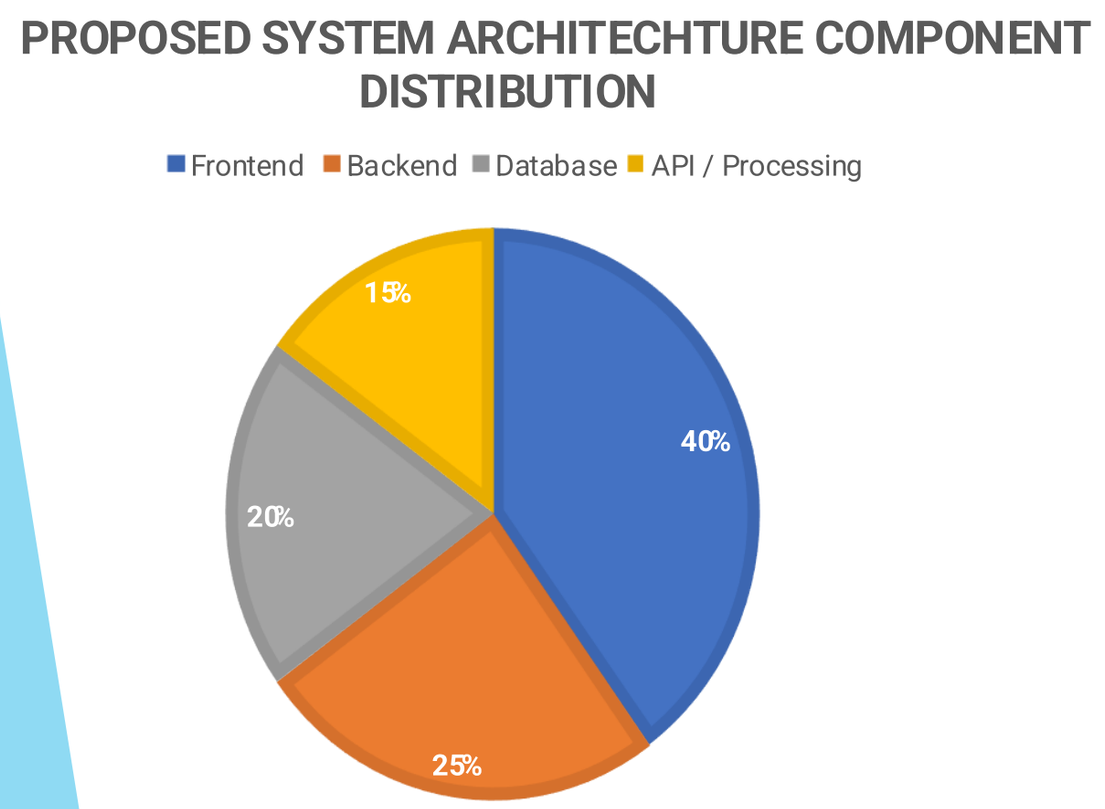
Thakur College Of Engineering And Technology — High — Mumba.

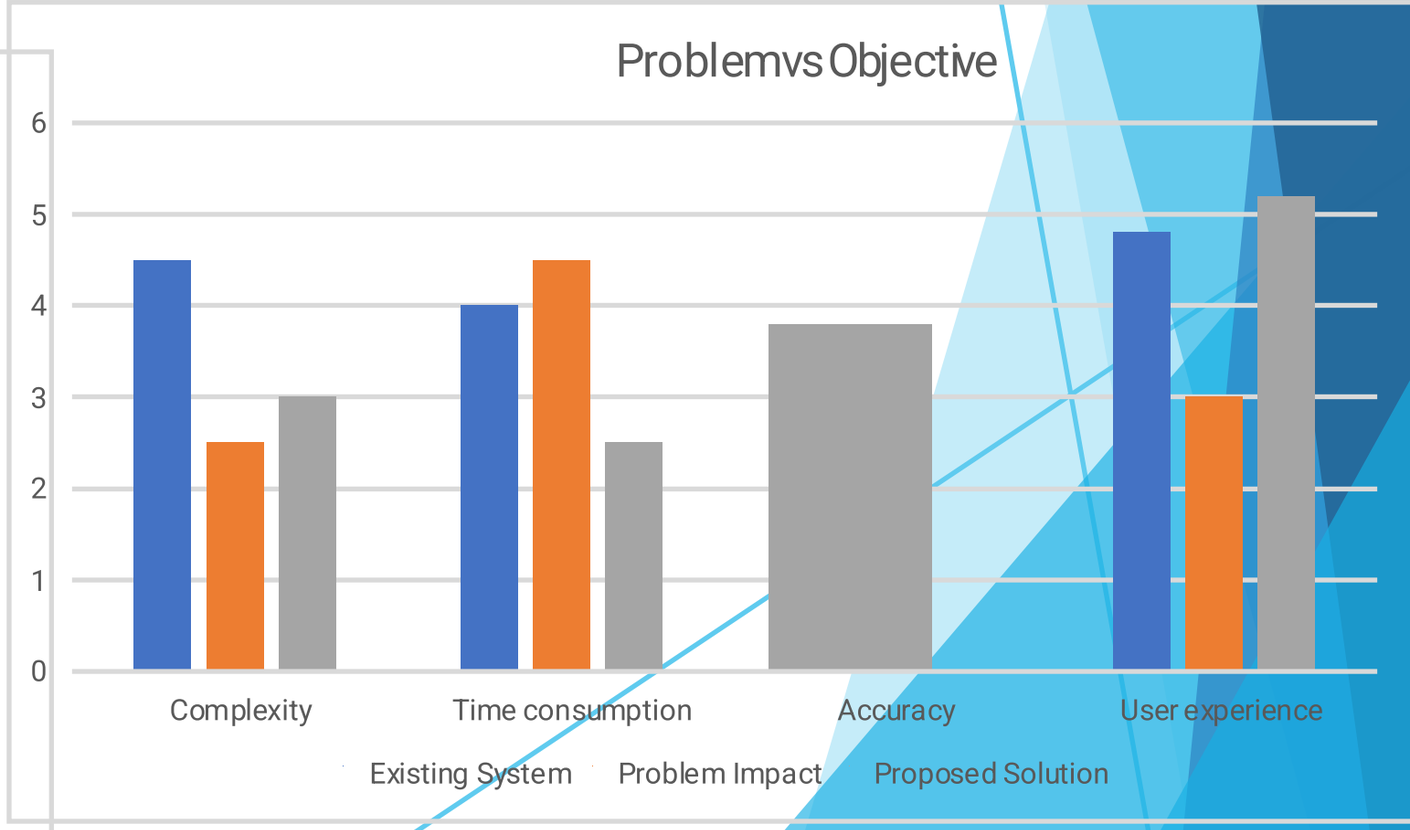
27°C | ENG | 10:43:03

Results & Analysis

Comparison with Existing Systems

Feature	Existing System	UNISCOUT System
Personalization	Low	High
Accuracy	~70%	**87%**
Real-time Data	Limited	Yes
Ease of Use	Moderate	High
Prediction Capability	Basic	Advanced





Conclusion and Future work

The **UNISCOUT College Recommendation System** successfully provides:

- Personalized college suggestions based on student inputs
- Data-driven analysis using cutoff trends
- The system helps students:
- Make informed decisions
- Reduce confusion during CAP rounds

Objectives Achieved:

  - Automated college filtering and recommendation
  - Improved accuracy in predicting admission chances
  - User-friendly and efficient interface

Future Work

- **Scope of Improvement:**
  - Integration of AI/ML models for more accurate predictions
  - Inclusion of additional parameters (placements, fees, reviews)
  - Real-time updates with latest CAP data
- Development of:
  - Mobile application for wider accessibility
  - Chatbot-based guidance system
- Expansion to:
  - Other entrance exams (JEE, NEET, etc.)
  - Nationwide college recommendation system

References

- MHT CET Cell, Government of Maharashtra, "Centralized Admission Process (CAP) Guidelines," 2024.
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- F. Ricci, L. Rokach, and B. Shapira, "Recommender Systems Handbook," *Springer*, 2015.
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Thank You...!!